

PQC in Practice: A 2026 Literature Review on Protocol-Level Transitions

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DAILY MONITORING

PQC in Practice: A 2026 Literature Review on Protocol-Level Transitions The year 2026 marks a critical juncture in the transition to Post-Quantum Cryptography (PQC). As quantum computers capable of breaking current cryptographic standards loom closer, the internet's security infrastructure is in a race to upgrade. A recent comprehensive literature review, "Post-Quantum Cryptography in Practice: A Literature Review of Protocol-Level Transitions and Readiness," synthesizes the global efforts, challenges, and state of readiness for integrating PQC into the core protocols that underpin our digital world. This study provides a crucial cross-protocol analysis, moving beyond isolated examinations of single technologies to offer a unified view of the migration landscape. It covers foundational protocols like TLS, SSH, and various VPN technologies, assessing their path toward quantum resistance. ### Key Findings: A Hybrid Future and Uneven Progress The review highlights several key themes. First and foremost is the widespread adoption of **hybrid cryptographic approaches**. To ensure a secure transition and mitigate risks from both classical and quantum adversaries, protocols are implementing a combination of a well-understood classical algorithm (like X25519 for key exchange) and a new PQC algorithm (like ML-KEM). An attacker would need to break both to compromise the connection, providing a robust defense-in-depth strategy. However, progress is not uniform across the protocol stack. The review reveals a significant disparity in adoption and standardization: - **Leaders:** Protocols like TLS 1.3 are at the forefront, with mature IETF drafts and multiple experimental and production deployments by major tech companies. The hybrid model has been well-defined and is seeing increasing adoption. - **In-Progress:** Technologies like SSH and IPsec (for VPNs) have standardized mechanisms for hybrid key exchange, but widespread production adoption lags behind. While the technical specifications exist, inertia and the complexity of updating diverse implementations present challenges. - **Laggards:** Protocols dealing with digital signatures, such as DNSSEC, face more fundamental hurdles. The larger signature sizes of PQC algorithms like ML-DSA and SLH-DSA conflict with protocol constraints, such as message size limits in UDP packets, creating significant structural barriers to adoption.

The Performance Challenge

A major focus of the review is the real-world implementation challenges. Beyond theoretical security, the performance of PQC algorithms—in terms of computational overhead, key and signature sizes, and handshake latency—is a critical factor. Early adopters have demonstrated that while the impact is manageable for protocols like TLS on modern hardware, it can be a significant concern for resource-constrained IoT devices or high-throughput network appliances.

Protocol Readiness Comparison The following table, derived from the literature review's findings, summarizes the PQC transition status for key internet protocols as of 2026.

Protocol	Primary Use Case	PQC Migration Status	Key Challenges
TLS 1.3	Web Security (HTTPS)	Leading	Broad server-side adoption, middlebox compatibility.
SSH	Secure Remote Admin	In Progress	Interoperability between different implementations, updating legacy clients/servers.
IPsec/IKEv2	Enterprise VPNs	In Progress	Performance overhead on network hardware, fragmentation with larger keys.
DNSSEC	DNS Integrity	Lagging	Large signature sizes conflicting with UDP packet size limits.

Applicability to Real-World Cybersecurity This comprehensive review is not merely an academic exercise; it's a vital tool for cybersecurity professionals. It underscores that the PQC transition is not a monolithic event but a complex, multi-year process with different timelines for various parts of the internet infrastructure. For security architects and network administrators, this means: 1. **Prioritizing Hybrid Deployments:** Organizations should prioritize deploying hybrid PQC solutions for public-facing services like web servers (TLS) and VPN gateways (IPsec) to protect against "Harvest Now, Decrypt Later" attacks. 2. **Vendor and Supply Chain Audits:** It is crucial to engage with software and hardware vendors to understand their PQC roadmaps. The readiness of the entire technology stack, from operating systems to network appliances, will determine the success of the migration. 3. **Preparing for Signature Scheme Transitions:** While key exchange is the immediate focus, the challenges highlighted in DNSSEC signal that migrating digital signatures will be

a more difficult, long-term project requiring careful planning. The findings confirm that while the path is complex, the internet engineering community is making tangible progress toward a quantum-resistant future. The lessons learned from the leaders in this transition will be invaluable for securing the entire digital ecosystem. --- ### Daily Quiz 1. What is the primary strategy being adopted by protocols like TLS 1.3 to transition to PQC securely? a) Immediately replacing all classical algorithms with PQC algorithms. b) Using a hybrid approach combining a classical and a PQC algorithm. c) Waiting for quantum computers to become a proven threat before migrating. 2. According to the 2026 literature review, which type of protocol faces the most significant structural barriers to PQC adoption? a) Key exchange protocols like TLS. b) Remote administration protocols like SSH. c) Digital signature protocols like DNSSEC. 3. What is a key practical challenge mentioned in the review regarding the real-world implementation of PQC? a) Lack of available PQC algorithms. b) Performance overhead, including larger key/signature sizes and latency. c) Resistance from the open-source community. --- *Answers: 1-b, 2-c, 3-b*